Length Extrapolation of Transformers: A Survey from the Perspective of Positional Encoding

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Abstract

Built upon the Transformer, large language models (LLMs) have captured worldwide attention due to their remarkable abilities. Nevertheless, all Transformer-based models including LLMs suffer from a preset length limit and can hardly generalize from short training sequences to longer inference ones, namely, they can not perform **length extrapolation** to handle long sequences. Thus, numerous methods have emerged to enhance the length extrapolation of Transformers. Despite the great research efforts, a systematic survey is still lacking. To fill this gap, we delve into these advances in a unified notation from the perspective of positional encoding (PE), as it has been considered the primary factor on length extrapolation. Specifically, we begin with extrapolatable PEs that have dominated this research field. Then, we dive into extrapolation methods based on them, covering position interpolation and randomized position methods. Finally, several challenges and future directions in this area are highlighted. Through this survey, We aim to enable the reader to gain a deep understanding of existing methods and provide stimuli for future research.

1 Introduction

It has been suggested that with limited learning resources, humans can potentially comprehend utterances of infinite length by understanding their components and structures (Chomsky, 1957; MONTAGUE, 1970). In natural language processing (NLP), given the limited training data (Kazemnejad et al., 2023) and compute, models can not learn from large-scale long sequences and thus are also expected to possess such generalization ability to process long sequences (Shaham et al., 2023). However, it is a challenging task for the de facto Transformer architecture (Vaswani et al., 2017), though Transformer-based large language models (LLMs) (Touvron et al., 2023a; OpenAI, 2023) have drastically advanced the NLP field.

Models based on the Transformer are trained on sequences with a maximum length (Raffel et al., 2020; Zhang et al., 2020; Brown et al., 2020), as a result of its quadratic memory and computational complexity with regard to input length. To make matters worse, some research reveals that Transformers might have gained their performance from surface-level memorization instead of abstract, generalizable skills (Razeghi et al., 2022; Wu et al., 2024), which means Transformers can hardly break through the maximum training length and perform poorly on sequences longer than seen during training (Dai et al., 2019; Neishi and Yoshinaga, 2019), i.e., they can not perform **length extrapolation**.

The length limit together with poor length extrapolation prevents LLMs from handling long sequences, such as DNA and protein sequences (Abramson et al., 2024), high-resolution images (Liu et al., 2023a), and even videos (Lin et al., 2023). Moreover, existing approaches for harnessing the full potential of LLMs also demand a larger context window, to incorporate elaborate prompts (Liu et al., 2023b), sufficient in-context demonstrations (Brown et al., 2020) and long-term memory of agents (Park et al., 2023). Hence, there is a growing body of research trying to strengthen length extrapolation of LLMs (Press et al., 2021; Ontanon et al., 2022; Anil et al., 2022; Chi et al., 2023b; Sun et al., 2023).

Despite the prosperity in this area, a systematic survey is still lacking. We aim to fill this blank by investigating existing approaches that enable and enhance length extrapolation of Transformers. Specifically, a brief formal introduction to Transformer is given in §2 as a solid foundation for further discussion. Then, we comprehensively summarize extrapolatable PEs proposed from the birth of Transformer to the prevalence of LLMs in §3. Note that We focus exclusively on PEs proposed for better extrapolation and omit others, since there

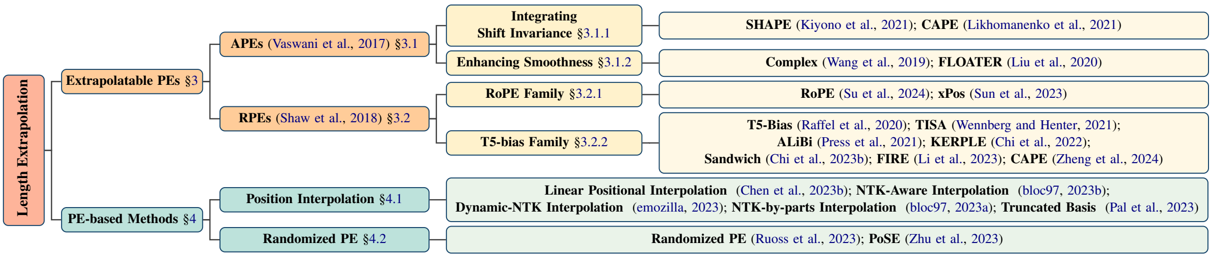

Figure 1: Taxonomy for length extrapolation of Transformers.

is already an intriguing survey on PEs of Transformer (Dufter et al., 2022). Based on these PEs, many novel methods emerge in the era of LLMs to further enhance extrapolation, which we intentionally centralize in §4, covering popular position interpolation methods and randomized methods. These advancements demonstrate the vibrancy and vastness of this area, from which we distill insights and future directions, represented in §5.

2 Preliminary

In this section, we follow Dufter et al. (2022) to present a formal description of the encoder layer of the Transformer, as the decoder layer is almost the same except for the cross-attention mechanism. Given an input matrix $\boldsymbol{X} \in \mathbb{R}^{n \times d}$ as a sequence of n embeddings with dimension d, an encoder layer $f : \mathbb{R}^{n \times d} \to \mathbb{R}^{n \times d}$ with $f(\boldsymbol{X}) = \boldsymbol{Z}$ is defined by:

$$\boldsymbol{C} = \frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d}} \tag{1}$$

$$\boldsymbol{A} = \mathrm{Softmax}(\boldsymbol{C})\boldsymbol{V} \tag{2}$$

$$\boldsymbol{O} = \mathrm{LayerNorm}_1(\boldsymbol{A} + \boldsymbol{X}) \tag{3}$$

$$\boldsymbol{F} = \mathrm{ReLU}(\boldsymbol{O}\boldsymbol{W}^{(f_1)} + \boldsymbol{b}^{(f_1)})\boldsymbol{W}^{(f_2)} + \boldsymbol{b}^{(f_2)} \tag{4}$$

$$\boldsymbol{Z} = \mathrm{LayerNorm}_2(\boldsymbol{O} + \boldsymbol{F}) \tag{5}$$

where $\boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}_q, \boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}_k, \boldsymbol{V} = \boldsymbol{X}\boldsymbol{W}_v$ are queries, keys and values, with $\boldsymbol{W}_q, \boldsymbol{W}_k, \boldsymbol{W}_v \in \mathbb{R}^{d \times d}$ being the projection matrices. Firstly, the compatibility scores $\boldsymbol{C}$ are computed as the dot product between queries and keys with a scaling factor[1] $1/\sqrt{d}$ (Equation 1). Then, the row-wise softmax function converts compatibility scores into weights, and the weighted sum of the values is the output of the attention layer (Equation 2). The fully connected feed-forward network consists of two linear transformations with a ReLU activation between (Equation 4), with parameters $\boldsymbol{W}^{(f_1)} \in \mathbb{R}^{d \times d_f}, \boldsymbol{W}^{(f_2)} \in \mathbb{R}^{d_f \times d}, \boldsymbol{b}^{f(1)} \in \mathbb{R}^{d_f}, \boldsymbol{b}^{(f_2)} \in \mathbb{R}^{d}$, where d_f is the intermediate dimension.

Note that in the above descriptions, we have not imposed any limit on input length n, which means the Transformer is naturally equipped with a notion of length extrapolation. Theoretically, a fixed setting of Transformer weights defines a sequence-to-sequence function on sequences of *arbitrary length* (Yun et al., 2019). If the function applies the correct transformation for inputs of any length, it is expected to length extrapolate (Zhou et al., 2023).

However, we have to break this nature by integrating PE with Transformers to inject position information into them. Otherwise, they are **permutation equivalent** or **order invariant**[2]. Thus, PEs are central to length extrapolation and form the core focus of this survey.

3 Extrapolatable Positional Encodings

Sinusoidal position embeddings are proposed with Transformer as it may help extrapolate to longer sequences beyond training (Vaswani et al., 2017). The idea behind this claim, that length extrapolation can be enabled by simply changing PE, has been widely supported and demonstrated (Neishi and Yoshinaga, 2019; Press et al., 2021; Ruoss et al., 2023). Hence, developing better PEs has been the predominant avenue to enhance length extrapolation of Transformers. Table 1 presents a characterization of these extrapolatable PEs.

Basically, absolute positional encodings (APEs) map each position to a unique representation and integrate it with corresponding word embedding, while relative positional encodings (RPEs) encode the relative distance between tokens and directly inject it into the attention module. Besides, RPEs usually keep modifications independent of value vectors and leaves them not entangled with position information. Hence, position information of

[1] We will omit this scaling factor in the following for simplicity and clarity.

[2] Note that some existing research suggests causal language models can learn position information without PE (Tsai et al., 2019; Haviv et al., 2022; Chi et al., 2023a).

	PE	Manifestation	Learnable	Integration	Injection Layer
	Sinusoidal (Vaswani et al., 2017)	Embedding	✗	Add	Initial
APE	*with Shift Invariance*				
	SHAPE (Kiyono et al., 2021)	Embedding	✗	Add	Initial
	CAPE (Likhomanenko et al., 2021)	Embedding	✗	Add	Initial
	with Smoothness				
	Complex (Wang et al., 2020)	Embedding	✓	Multiply	Initial
	FLOATER (Liu et al., 2020)	Embedding	✓	Add	Initial
	Shaw et al. (2018)	Embedding	✓	Add	Every
RPE	*T5 Family*				
	T5 Bias (Raffel et al., 2020)	Bias	✓	Add	Every
	ALiBi (Press et al., 2021)	Bias	✗	Add	Every
	KERPLE (Chi et al., 2022)	Bias	✓	Add	Every
	SANDWICH (Chi et al., 2023b)	Embedding	✗	Add	Every
	FIRE (Li et al., 2023)	Bias	✓	Add	Every
	CAPE (Zheng et al., 2024)	Bias	✓	Add	Every
	RoPE Family				
	RoPE (Su et al., 2024)	Embedding	✗	Multiply	Every
	xPOS (Sun et al., 2023)	Embedding	✗	Multiply	Every

Table 1: A list of extrapolatable PEs. **Bolded** methods are proposed or widely adopted for LLMs. *Manifestation* shows how the position infomation is introduced. *Learnable* shows whether it can adjust based on the input. *Integration* shows how the position representations are integrated with token representations. *Injection Layer* shows the injecting position PE.

RPEs can be scalars and usually recurs at each layer. Figure 2 illustrates these general differences. We divide Table 1 and this section based on whether the PE is absolute or relative, as existing research suggests this distinction significantly impacts length extrapolation (Neishi and Yoshinaga, 2019; Likhomanenko et al., 2021; Chi et al., 2022).

3.1 Absolute Positional Encodings

Specifically, for a token in position pos, the sinusoidal position embedding is defined as:

$$[\dots, \sin(\frac{pos}{10000^{2i/d}}), \cos(\frac{pos}{10000^{2i/d}}), \dots], \quad (6)$$

where $i \in [0, d/2-1]$ is the dimension of the position embedding and d denotes model dimension. Then, each position embedding is added to the corresponding token embedding and the sum is fed into Transformer, so the compatibility score between query $\boldsymbol{q}_i$ and key $\boldsymbol{k}_j$ can be formalized as

$$\boldsymbol{q}_i\boldsymbol{k}_j^T = ((\boldsymbol{x}_i + \boldsymbol{p}_i)\boldsymbol{W}_q)((\boldsymbol{x}_j + \boldsymbol{p}_j)\boldsymbol{W}_k)^T. \quad (7)$$

This equation is the basis of many other PEs.

However, researchers subsequently found that sinusoidal APE is hard to extrapolate (Dai et al., 2019; Neishi and Yoshinaga, 2019). Hence, a wide variety of APEs have been proposed to enhance sinusoidal APE and extrapolation of Transformers from different perspectives, either trying to integrate shift invariance in sinusoidal APE (§3.1.1) or aiming to generate position embeddings varying smoothly with position indices (§3.1.2).

3.1.1 Integrating Shift Invariance

Taking inspiration from the three properties of PEs proposed by Wang et al. (2020), Kiyono et al. (2021) speculated superior extrapolation performance comes from *shift invariance*, the property of a function to not change its output even if its input is shifted. Aiming to incorporate the benefit of shift invariance in sinusoidal APE, they simply shift every position index of a sequence by a random offset k, which prevents the model from using absolute positions and instead encourages the use of relative positions.

Following a similar idea, Likhomanenko et al. (2021) took it a step further by leveraging continuous signals. In addition to shifting every position index of APE by an identical random offset, which they call *global shift*, they also introduced *local shift*, i.e., shifting each position index by a different random shift, and *global scaling*, i.e., scaling every position index by an identical random scalar, to further prevent capturing spontaneous correlations and memorizing distances.

3.1.2 Enhancing Smoothness

Apart from above relatively straightforward methods based on sinusoidal APE, there are several APEs taking quite different theoretical avenues to enhance length extrapolation, aiming to improve the smoothness of the position representations.

Wang et al. (2019) proposed to extend each word embedding as a continuous function over an independent variable, i.e., position, so that word representations vary smoothly with increasing positions. Through mathematically sound derivation, their general complex-valued embedding $f(j, pos)$ of a word w_j in position pos is

$$[r_{j,1}e^{i(\omega_{j,1}pos+\theta_{j,1})}, \cdots, r_{j,d}e^{i(\omega_{j,d}pos+\theta_{j,d})}], \quad (8)$$

where amplitude $\boldsymbol{r} = [r_{j,1}, \dots, r_{j,d}]$, frequency $\boldsymbol{\omega} = [\omega_{j,1}, \dots, \omega_{j,d}]$ and initial phrase $\boldsymbol{\theta} = [\theta_{j,1}, \dots, \theta_{j,d}]$ are all trainable. In addition to representing positions in complex plane for the first time, multiplying position embeddings with word embeddings is another of their innovations.

An alternative approach is to directly capture the dynamics between position representations. Liu et al. (2020) introduced a dynamical system to model position representations $\{\boldsymbol{p}_i \in \mathbb{R}^d : i =$

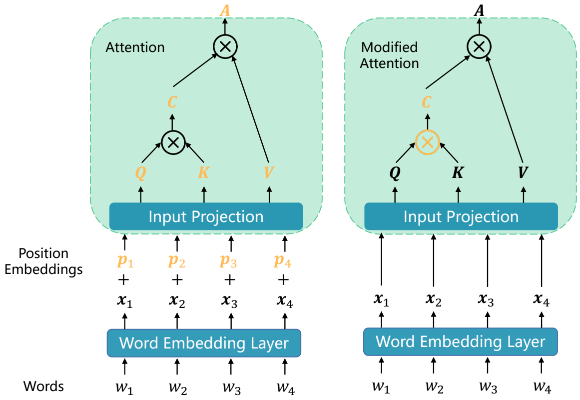

Figure 2: General differences between APE (left part) and RPE (right part), where orange denotes elements holding position information.

$1, \ldots, n\}$, which can be characterized as

$$\boldsymbol{p}(t)=\boldsymbol{p}(s)+\int_s^t \boldsymbol{h}(\tau, \boldsymbol{p}(\tau) ; \boldsymbol{\theta}_h) d \tau, 0 \leq s \leq t<\infty \tag{9}$$

with an initial vector $\boldsymbol{p}(0)$, where $\boldsymbol{p}(t): \mathbb{R}_{+} \mapsto \mathbb{R}^d$ is the continuous version of the discrete sequences $\{\boldsymbol{p}_i\}$. $\boldsymbol{h}(\tau, \boldsymbol{p}(\tau) ; \boldsymbol{\theta}_h)$, which is the "latent force" that drives the changes from $\boldsymbol{p}_i$ to $\boldsymbol{p}_{i+1}$, is actually a neural network parameterized by $\boldsymbol{\theta}_h$ and takes in the previous state $(\tau, \boldsymbol{p}(\tau))$.

Highlights: As the first PE for Transformer, sinusoidal APE has a significant impact on PEs thereafter, despite its poor extrapolation. To improve this, researchers either leverage random shift to incorporate shift invariance in sinusoidal APE or generate position embeddings varying smoothly with position. Among them, simple random shifting is like a small patch for sinusoidal APE and has limited benefits for extrapolation, at the cost of possible semantic confusion in position encoding, while the latter can hopefully lead to better extrapolation, coming with a much higher parameter- and computation-complexity.

3.2 Relative Positional Encodings

Albeit for the efforts in extrapolatable APEs, it is believed that RPEs are theoretically capable of running on unseen lengths and are more robust to input length change (Neishi and Yoshinaga, 2019; Likhomanenko et al., 2021; Chi et al., 2022), as RPEs only rely on relative position information, which means they encode the idea of shift invariance naturally and are not subject to a maximum position value. Besides, there is a consensus that in natural language, it is not absolute but relative position that matters (Huang et al., 2020; Sinha et al., 2022). Thus, RPEs become the dominant way to encode positions, which we detail in this section. Before that, we reformulate Equation 7 as follows to clarify the perspective of RPEs:

$$\boldsymbol{q}_i \boldsymbol{k}_j^T=(\boldsymbol{x}_i \boldsymbol{W}_q)(\boldsymbol{x}_j \boldsymbol{W}_k)^T \oplus \boldsymbol{p}(j-i), \tag{10}$$

where $\boldsymbol{p}(j-i)$ encodes the relative position information, $\oplus$ denotes any approach of integrating the position information into the compatibility score.

Among the first, Shaw et al. (2018) introduced the idea of RPE based on above formulation. Specifically, they concretized Equation 10 as

$$\boldsymbol{q}_i \boldsymbol{k}_j^T=(\boldsymbol{x}_i \boldsymbol{W}_q)(\boldsymbol{x}_j \boldsymbol{W}_k+\boldsymbol{p}_r)^T, \tag{11}$$

where $\boldsymbol{p}_r \in \mathbb{R}^d$ is a trainable relative position embedding and $r=\operatorname{clip}(j-i, r_{\min }, r_{\max })$ denotes the clipped relative position. By clipping the relative positions to a determined range, the number of position embeddings to be learned is reduced and length extrapolation is enhanced as unseen position embeddings are avoided. This RPE can also be regarded as a derivation of sinusoidal APE. Following this line, more RPEs have been proposed to better model position information, such as Dai et al. (2019), Huang et al. (2020) and TUPE (Ke et al., 2020). We omit them here since they are not proposed for stronger length extrapolation.

3.2.1 RoPE Family

Also inspired by sinusoidal APE, Su et al. (2024) proposed to multiply keys and queries by rotation matrices, leaving compatibility scores as

$$\begin{aligned}\boldsymbol{q}_i^T \boldsymbol{k}_j &=(\boldsymbol{R}_{\Theta, i}^d \boldsymbol{x}_i \boldsymbol{W}_q)^T(\boldsymbol{R}_{\Theta, j}^d \boldsymbol{x}_j \boldsymbol{W}_k) \\ &=\boldsymbol{W}_q^T \boldsymbol{x}_i^T \boldsymbol{R}_{\Theta, j-i}^d \boldsymbol{x}_j \boldsymbol{W}_k,\end{aligned} \tag{12}$$

where $\boldsymbol{R}_{\Theta, j-i}^d=(\boldsymbol{R}_{\Theta, i}^d)^T \boldsymbol{R}_{\Theta, j}^d$ with $\boldsymbol{R}_{\Theta, i}^d$ being a block-diagonal matrix with rotation matrices

$$\begin{pmatrix}\cos i \theta_m & -\sin i \theta_m \\ \sin i \theta_m & \cos i \theta_m\end{pmatrix} \tag{13}$$

on its diagonal, given the parameters $\Theta=(\theta_m)_{m=1,2, \ldots, d / 2}$ where $\theta_m=10000^{-2(m-1) / d}$. Here the *base* is 10000, and $\lambda_m=2 \pi / \theta_m$ is wavelength. This method is called Rotary Position Embedding (RoPE) as intuitively it rotates key/value embeddings according to their position index:

$$f_{\{q, k\}}(\boldsymbol{x}_i, i)=\boldsymbol{R}_{\Theta, i}^d \boldsymbol{x}_i \boldsymbol{W}_{\{q, k\}}. \tag{14}$$

It is noteworthy that despite the absolute nature of this rotary process, the compatibility score and thus attention depend only on relative distance. This property together with long-term decay for inter-token product benefit length extrapolation.

As RoPE has been widely used in popular LLMs (Touvron et al., 2023a; Jiang et al., 2023; Anil et al., 2023), there are some variants proposed to improve it. Sun et al. (2023) defined attention score expectation between two tokens at a specific distance and further attributed the poor extrapolation of RoPE to the dramatic oscillation of their attention expectations. They proposed to fix this issue by incorporating a balancing term to punish the oscillation of unstable dimensions and keep the distribution of stable ones, which can be simplified to:

$$\boldsymbol{q}_i^T \boldsymbol{k}_j = \gamma^{i-j} \boldsymbol{W}_q^T \boldsymbol{x}_i^T \boldsymbol{R}_{\Theta, j-i}^d \boldsymbol{x}_j \boldsymbol{W}_k, \tag{15}$$

where $\gamma \in (0, 1)$ is a scalar hyperparameter.

3.2.2 T5-Bias Family

Different from complex embedding form, some researchers reduce position information $\boldsymbol{p}(j-i)$ to a simpler form. Raffel et al. (2020) utilized learnable scalars to represent relative position information:

$$\boldsymbol{q}_i \boldsymbol{k}_j^T = (\boldsymbol{x}_i \boldsymbol{W}_q)(\boldsymbol{x}_j \boldsymbol{W}_k)^T + \beta_{i,j}. \tag{16}$$

In addition, they extended the clipping mechanism by a logarithmic bucket assignment to achieve precise discrimination of nearby positions and less precise discrimination of further positions, which further reduces the parameters to be learned and is beneficial for extrapolation (Chi et al., 2022). Moreover, Wennberg and Henter (2021) introduced TISE, which leverages a radial-basis function of relative distance with multiple trainable parameters to add a bias to attention scores.

As the first PE aiming mainly for length extrapolation, ALiBi (Press et al., 2021) takes an even simpler way to represent relative position:

$$\boldsymbol{q}_i \boldsymbol{k}_j^T = (\boldsymbol{x}_i \boldsymbol{W}_q)(\boldsymbol{x}_j \boldsymbol{W}_k)^T + m(j-i), \tag{17}$$

where scalar m is a head-specific slope fixed before training. It is worth noting that there is no additional learnable parameter, which leads to superior efficiency and may also contribute to better extrapolation of ALiBi. Empirical experiments on language modeling demonstrated its superiority.

From the perspective of kernel methods, Chi et al. (2022) considered ALiBi as a triangle kernel and extended it to KERPLE, a framework that generalizes relative position embeddings for extrapolation by kernelizing positional differences using conditionally positive definite kernels. In this framework, various RPEs can be derived from different conditionally positive definite kernels in a principled way, among which the logarithmic variant achieves preferred extrapolation performance, by calculating the compatibility score as follows:

$$\boldsymbol{q}_i^T \boldsymbol{k}_j = (\boldsymbol{x}_i \boldsymbol{W}_q)^T (\boldsymbol{x}_j \boldsymbol{W}_k) - r_1 \cdot \log(1 + r_2 |i-j|), \tag{18}$$

where r_1, r_2 are positive scalar parameters.

Aware of the overfitting issue of sinusoidal APE, Chi et al. (2023b) proposed to overcome it by simplifying sinusoidal APE to a new RPE, Sandwich. Specifically, they dropped the cross terms in Equation 7 and kept the inner product of two position embeddings as position information:

$$\boldsymbol{q}_i^T \boldsymbol{k}_j = (\boldsymbol{x}_i \boldsymbol{w}_q)^T (\boldsymbol{x}_j \boldsymbol{W}_k) + \boldsymbol{p}_i^T \boldsymbol{p}_j. \tag{19}$$

It is worth noting that in this formula, $\boldsymbol{p}_i^T \boldsymbol{p}_j$ becomes a temporal bias term with the same decay-with-distance pattern as ALiBi, which is exactly what the authors want to achieve as they suggested this pattern is likely to be the secret to successful length extrapolation. Besides, since position embeddings here only need to interact with themselves, the authors make the dimension of them a hyperparameter to further improve performance.

FIRE (Li et al., 2023) integrates positional information into Transformers following T5 bias:

$$\boldsymbol{q}_i \boldsymbol{k}_j^T = (\boldsymbol{x}_i \boldsymbol{W}_q)(\boldsymbol{x}_j \boldsymbol{W}_k)^T + b(i,j), \tag{20}$$

where the bias $b(i,j)$ is mapped from positions using a learnable continuous function $f_\theta : \mathbb{R} \to \mathbb{R}$, e.g., MLP. To avoid the generalization issue when the inputs are outside the training domain of the function, they proposed progressive interpolation by normalizing the distance by query position index, namely $b(i,j) = f_\theta(\frac{i-j}{i})$. Note that in causal attention, the normalized distance is always bounded between $[0, 1]$, which aligns the inference domain with the training domain for any sequence lengths, leading to better length extrapolation.

However, the above methods separate positional bias from semantics completely, which may cause semantic similarity to be overshadowed by position information. Hence, Zheng et al. (2024) proposed Context-Adaptive Positional Encoding (CAPE) to integrate both semantic and positional information:

$$\begin{aligned} \boldsymbol{q}_i \boldsymbol{k}_j{}^T &= (\boldsymbol{x}_i \boldsymbol{W}_q)(\boldsymbol{x}_j \boldsymbol{W}_k)^T \\ &+ f((\boldsymbol{x}_i \boldsymbol{W}_q)(\boldsymbol{x}_j \boldsymbol{W}_k)^T, b(i,j)). \end{aligned} \tag{21}$$

Here $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is parameterized by a two-layer LeakyReLU neural network and $b(i, j)$ come from other RPEs(e.g., ALiBi and FIRE).

In addition to RPEs introduced previously, there are some methods can not be categorized into RoPE or T5-bias family. He et al. (2024) introduce bilevel PE that employs two distinct PE for each position: an APE for intra-segment position to help model capture the semantics contained therein, while an RPE for inter-segment position to capture relationships between segments and exhibits extrapolation. This decoupling offers greater flexibility in addressing the length extrapolation problem.

Based on the observation that existing PEs use token as the unit of measurement, Golovneva et al. (2024) claimed that this feature prevents PEs from generalizing to higher levels of abstraction such as sentences and paragraphs. Therefore, they proposed Contextual Positional Encoding (CoPE), which allows the model to determine semantic unit (e.g., word and sentence) and assign tokens therein a same position index. Since CoPE can distribute positions to a much larger number of tokens and focus attention on semantic units at a higher level of abstraction, it exhibits stronger extrapolation.

Highlights: Earlier RPEs had been greatly influenced by sinusoidal APEs by modifying terms in Equation 7 and replacing absolute embeddings with relative embeddings. These methods usually leverage clipping or binning strategy to avoid out-of-distribution position embeddings and enhance extrapolation. Since RPEs decouple the one-to-one correspondence between position and position representation, incorporating bias term directly into compatibility score (Equation 10) becomes a feasible and even better way to encode positional information, which is much simpler and naturally disentangles value vectors and position information. However, despite the strong extrapolation of these bias methods, they can not represent complex distance-attention functions based on Fourier basis like RoPE. Therefore, RoPE become the de facto PE of recent LLMs due to its advanced general performance, in spite of its poor extrapolation.

4 Extrapolation Methods in LLMs Era

Based on PEs in §3, various methods have been developed to further enhance length extrapolation of LLMs. This section is separated in response to this wave, focusing on interpolation methods and randomized PEs, as illustrated in Figure 3.

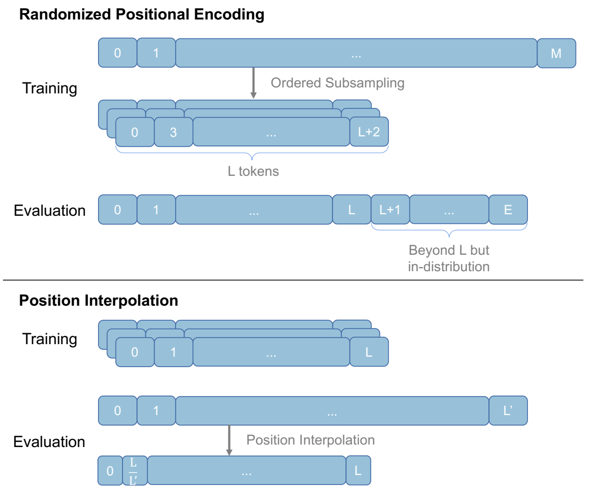

Figure 3: Essentials of position interpolation and randomized PE. Randomized PE aims to ensure that positions falling outside the context window at inference remain in distribution through advanced exposure in training. Position interpolation, on the other hand, works during the inference stage by scaling a longer position range into the original context window.

4.1 Position Interpolation

Despite the large quantity of PEs with better extrapolation, RoPE has been most widely adopted in recent LLMs due to its superior in-distribution performance. Hence, loads of methods have been proposed to enhance the extrapolation of RoPE, the most prevalent of which is position interpolation.

Chen et al. (2023b) firstly [3] introduced position interpolation for RoPE to extrapolate LLMs to longer sequences by applying linear scaling to down-scale position indices so that the maximum position index matches the previous length limit during pre-training. Formally, this method replaces RoPE f (Equation 14) by f' defined as $f'(\boldsymbol{x}, i) = f(\boldsymbol{x}, \frac{iL}{L'})$, where L is the length limit during pre-training and L' is the longer sequence length at inference. The *scale ratio* $\kappa = L'/L$ transforms position n to n/κ. This method reduces absolute position indices from $[0, L')$ to $[0, L)$ and maximum relative distance from L' to L, aligning the ranges of position indices and relative distances to mitigate effects on attention score computation.

However, from the perspective of Neural Tangent Kernel (NTK) theory (Jacot et al., 2018), simply interpolating RoPE's Fourier space linearly will cause the loss of high-frequency information and prevent models from distinguishing nearby po-

[3]There is a concurrent work: `https://kaiokendev.github.io/til#extending-context-to-8k`

sitions.Hence, NTK-Aware Scaled RoPE (NTK-aware interpolation) (bloc97, 2023b) has been proposed by modifying the base of RoPE:

$$\theta^*_m = (b \cdot \kappa^{\frac{d}{d-2}})^{-2(m-1)/d}, \tag{22}$$

where b is the original base and κ is still the scale ratio. The core idea here is to scale high frequencies less and low frequencies more to reduce information loss of high frequencies. As NTK-aware interpolation does not scale the Fourier features directly, all positions are distinguishable from each other. Moreover, this method does not require any fine-tuning to extend the context window.

Further, Dynamic-NTK interpolation (emozilla, 2023) combined NTK-aware interpolation with dynamic scaling, using exact positions for tokens within pre-trained context window to prevent performance degradation and dynamically increases scale ratio κ as current sequence length increases to adjust positions beyond the window:

$$\kappa = \begin{cases} L'/L, & \text{if } L'/L > 1, \\ 1, & \text{otherwise}, \end{cases} \tag{23}$$

where L' is the sequence length of the current sequence, which will increase after each step.

Either scaling position indices or modifying bases, all position representations become closer to each other, impairing LLM's ability to distinguish the positional order of close-by tokens. Besides, bloc97 (2023a) observed that some RoPE dimensions have wavelengths longer than the pre-trained context window, where they presume absolute positional information remains intact[4]. Hence, they proposed NTK-by-parts, which does not interpolate dimensions of small wavelengths at all while always interpolating those of big ones.

Similar observations with NTK-by-parts have been made by Pal et al. (2023), based on which they proposed to use the truncated basis:

$$\theta^*_i = \begin{cases} \theta_i & \text{for } \theta_i \geq b, \\ \rho & \text{for } a < \theta_i < b, \\ 0 & \text{for } \theta_i < a. \end{cases} \tag{24}$$

where ρ is a fixed value that is relatively small, and a and b are chosen cutoff values. This way, models will experience all values of the basis in the context length used during fine-tuning by choosing appropriate cutoff values, and are supposed to extrapolate better during inference.

Additionally, Peng et al. (2023) observed that by introducing a temperature t into compatibility score before Softmax, perplexity decreases consistently. Combining this finding with NTK-by-parts interpolation, they subsequently proposed YaRN that surpasses previous interpolation methods in both fine-tuned and non-fine-tuned scenarios.

The interpolation methods reflect the critical impact of the rotary base of RoPE on length extrapolation, prompting efforts to enhance extrapolation of RoPE-based LLM by fine-tuning it with a scaled base (Xiong et al., 2023; Rozière et al., 2023; Liu et al., 2023c). However, fixed scaling factors overlook the gradual length-extension process and impair performance at shorter lengths, leading to the proposal of dynamic scaling methods (Chen et al., 2023a; Zhang et al., 2024; Ding et al., 2024). Innovatively, Wang et al. (2024) scale each dimension's base by rounding its wavelength to the nearest integer, avoiding phase shifts after each full rotation.

Highlights: Recently, position interpolation methods have raised widespread interest in the research community, as a natural result of their superior extrapolation performance and extremely low overhead. Current interpolation methods either interpolate position indices or RoPE's base, guided by sound theoretical intuition. Besides, different from other extrapolation methods, position interpolation methods have already seen their presence in the open-source models (Bai et al., 2023; Touvron et al., 2023b; AI et al., 2024).

4.2 Randomized Positional Encoding

For APEs and RPEs without a clipping mechanism, length extrapolation results in out-of-distribution position representations and performance degradation. To address this, randomized PEs are proposed to enable models to observe all possible position representations during training.

As a realization of this idea, Ruoss et al. (2023) proposed to simulate a much longer range of positions and randomly selects an ordered subset to fit the training context window for each iteration. Thus, through adequate training, we can ensure that the model encounters enough unique positions and all M positions have been fully trained, leading to consistent extrapolation performance.

For the same motives, PoSE (Zhu et al., 2023) also tries to simulate much longer inputs by manip-

[4]From the perspective of frequency, the full range of high-frequency components have been seen by the model during training, while low-frequency components have not. Thus, every position within the context window leads to a unique value in these low-frequency components, based on which models can determine the absolute position of each token.

ulating position indices within a fixed pre-trained context window. However, different from Ruoss et al. (2023), PoSE partitions the original sequence into several chunks and adjusts the position indices by adding a distinct skipping bias term between each chunk. This way, PoSE keeps the positions continuous in each chunk, which bears a close resemblance to pre-training, while at the same time allows the model to adapt to all positions within a longer context window.

Highlights: Essentially, randomized PEs simply decouple the pre-trained context window with longer inference length by introducing randomized positions during training, which boosts exposure of all positions in the longer context window. This idea is quite different from that of position interpolation methods, where the latter tries to interpolate positions during inference to make them fall into the trained range. For the same reason, position interpolation methods are mostly plug-and-play while randomized PEs usually need further fine-tuning, which makes position interpolation much more appealing due to its low overhead.

5 Discussion and Future Directions

Evaluation and Benchmark. Initially, researchers evaluated length extrapolation by training models on sequences with a length limit and testing them on slightly longer sequences (Liu et al., 2020; Likhomanenko et al., 2021). During this phase, evaluation samples and metrics came from various downstream tasks such as machine translation and question answering. As pre-trained language models proved versatile, making it easy to translate other NLP tasks into language modeling (Raffel et al., 2020; Brown et al., 2020), language modeling and perplexity became the standard for evaluating length extrapolation (Press et al., 2021; Haviv et al., 2022). However, it has become clear that perplexity alone does not adequately reflect downstream task performance and is insufficient (Tay et al., 2021; Kazemnejad et al., 2023; Pal et al., 2023; Hu et al., 2024). Therefore, dedicated benchmarks and evaluation methods are needed to further advance the field of length extrapolation.

Explainability and Principle. Despite remarkable progress, our understanding of length extrapolation remains limited, lacking a solid theoretical foundation. The decaying-with-distance pattern was initially thought to be crucial for extrapolatable PEs (Press et al., 2021; Su et al., 2024), But it was later shown to merely accommodate the recency bias of language modeling (Chi et al., 2023c). Besides, extrapolation methods usually avoid out-of-distribution positions via interpolation or advanced exposure. Thus, it remains unclear when or if Transformers length extrapolate in real-world scenarios (Zhou et al., 2023) and whether or how existing methods help with it.

Analysis on Synthetic Tasks. There is some research on synthetic tasks such as arithmetic and deductive reasoning, analyzing length generalization behavior of Transformers within small context window. One common observation is that Transformers often struggle with length extrapolation, whether they are trained from scratch on synthetic tasks (Lee et al., 2023; Kazemnejad et al., 2023), fine-tuned from pre-trained LLMs (Anil et al., 2022) or used in in-context learning (Saparov et al., 2023). As explanations, Dziri et al. (2023) hypothesize certain tasks may not possess the inherent compositionality and allow for shortcut pattern matching. On the other side, Transformers are proven to length extrapolate on specific tasks (Zhou et al., 2023; Xiao and Liu, 2024) or with the right combination of data format and PE (Zhou et al., 2024). Meanwhile, some studies show other factors in length extrapolation. Anil et al. (2022) find that fine-tuning regime, scaling data, model sizes, and compute does not improve length extrapolation, while scratchpad (Nye et al., 2022) or chain-of-thought (Wei et al., 2022) in the in-context learning regime do. In addition, Kazemnejad et al. (2023) show that Transformers without PE outperform explicit PEs including ALiBi and RoPE on small-scale arithmetic tasks. These studies have deepened our understanding of extrapolation and broadened our perspectives to go beyond PE, demonstrating that the extrapolation ability needs a systematic design where PE is crucial but by no means the sole component.

6 Conclusion

Through this survey, we systematically summarized existing methods and recent advances in length extrapolation from the perspective of PE. Specifically, we meticulously categorize extrapolatable PEs and further dive into methods based on these PEs in LLMs era. In addition, we highlight existing challenges and identify new trends in this research field, hoping to facilitate researchers and provide stimuli for future research.

Limitation

This survey presented a systematic review of existing methods and recent trends in length extrapolation of Transformers. However, due to the lack of standardized benchmark and evaluation methods, we primarily focus on high-level comparisons and distinctions in principle of different approaches, rather than fine-grained empirical analysis. Furthermore, in this work, we focus on length extrapolation studies aimed at extending the context window of LLMs in real-world scenarios. Although we acknowledge the importance of studies analyzing length generalization in synthetic tasks within a small context window as well, we provide only a brief discussion on them due to the page limitation.

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